OSCILLATOR + MULTIVIBRATOR CIRCUITS

Experiment 8 | EC3091D - Electronic Circuits Lab II

COMPLETE FRIEND-FOLLOWABLE SIMULATION, CALCULATIONS, OBSERVATIONS & PRINTING GUIDE

What this document provides

- Every circuit in the supplied laboratory handout: Wien-bridge oscillator, astable, monostable and bistable multivibrators.
- Three planned resistor/capacitor trials per circuit (12 captures in total), satisfying the instructor's request for 2-3 waveforms per circuit.
- Worked theoretical values, blank actual-result tables, methods for period/frequency/duty cycle/pulse width, and labeled A4 waveform attachment spaces.
- White-background *reference illustrations* plus instructions for obtaining authentic white-background LTspice waveform printouts.

Important evidence status

No open LTspice project or measured .raw data was provided with the PDF in this chat. Therefore, every graph labelled THEORETICAL or SCHEMATIC in this report is only a reference illustration. Actual simulator measurements and the 12 requested authentic waveform screenshots must be completed on the friend's computer. DO NOT submit reference illustrations as measured simulation results.

What the teacher requires

Print the simulation waveforms and stick them in the FAIR RECORD. Write the experiment in the ROUGH RECORD. For each circuit, change resistor and/or capacitor values to obtain 2-3 waveforms and note the relevant duty cycle, time period, frequency etc. (Use "N/A" with reason for measurements that do not apply.)

Deliverable	Quantity / purpose
Actual LTspice waveform printouts	12 target; 3 from each circuit
Record of actual component values	12 trial rows + supply and model
Cursors / measured quantities	Period, frequency; plus duty or pulse width where meaningful
Theoretical calculations	Shown in this report; verify against simulation
Fair record / rough record	Printouts / written experiment, respectively

Source: Supplied "Experiment-8 Oscillator and Multivibrator Circuits" lab PDF (3 pages); teacher message quoted above. Calculations are analysis added for this guide, not numbers printed in the lab sheet.

01 | THE EXACT WORK PLAN

Folder and file naming

Create a folder Lab8_Experiment8 with subfolders: 01_Wien, 02_Astable, 03_Monostable, 04_Bistable, 05_Printouts. Save each trial's .asc file, screenshot and observation notes. Do not overwrite the working baseline circuit.

Circuit	Trial names	Must appear in each plot
Wien bridge	W1, W2, W3	Output sinusoid, axes, R/C values, measured T and f
Astable	A1, A2, A3	V(out) square + optional V(C); measured T, f, T_HIGH, duty
Monostable	M1, M2, M3	V(in trigger) and V(out); pulse width, trigger period and rate
Bistable	B1, B2, B3	V(in trigger) and V(out); HIGH/LOW held states, edges

Repeat this sequence for every trial

01. Open the relevant original circuit or create it from the handout. Save As a separate uniquely named trial .asc.

02. Set R/C values from its planned-trial table; keep remaining components, power rails, LM741 model, diode direction and input setup unchanged unless a legitimate issue requires a documented correction.

03. Run transient simulation, select relevant nodes, zoom to 2-5 steady cycles or 2-3 distinct triggering events (never crop off axis scales).

04. Read crossing times with two waveform cursors. Record actual values and calculate f or pulse duty as appropriate.

05. Set the waveform background to WHITE, trace to DARK, and save a legible capture / print to PDF. Verify labels/axes are readable on paper.

06. Write one line of values in the observation table. Repeat for the next trial.

Final hand-in split

- ROUGH RECORD: aim, required apparatus/software, original circuit diagrams, theoretical formulas, component trial values, procedure, measured observation tables, calculations, result, and interpretation.
- FAIR RECORD: paste 2-3 actual LTspice waveform printouts for EACH of the 4 circuits, neatly label Trial #, R/C values, T, f and duty/pulse width or N/A.
- Maintain a local folder of .asc/.plt and evidence screenshots in case a plot needs to be reproduced.

For monostable the repetition frequency depends on external trigger rate; for bistable no autonomous period/frequency exists. Do not invent those entries.

02 | LTspice: CAPTURE REAL RESULTS

Basic operation - repeat as needed

01. Open *.asc in LTspice. If the 741 is a supplied subcircuit, keep its .include/.lib and required symbol pin mapping. Confirm +/-15 V rails for the multivibrator figures.

02. Right-click R/C to edit values. LTspice accepts 10k, 0.1u, 0.22u, 47n, 1Meg. Save As W1.asc / A1.asc etc.

03. Set a transient analysis suitable for the circuit: Wien 60-100 ms at 159 Hz, Astable 20-40 ms at around 1 kHz, Monostable 40 ms for 10 ms-spaced triggers, Bistable 30 ms for distinct edges. Extend if fewer than 3 transitions occur.

04. Click Run, then click the actual wire labelled OUT to plot V(out). For astable also click capacitor node VC; for triggered circuits add V(in) and V(out).

05. Use two waveform cursors on the output trace (click trace label, choose both cursors from right-click options if required). Place cursors on equivalent consecutive transitions or sine positive-going zero crossings.

06. Read delta t (= T) and its reciprocal (= f); for HIGH duration put the pair on rising and falling crossings. Log measured values, not theoretical estimates.

07. If trace looks flat/railed or does not toggle, investigate and FIX circuit/startup/trigger; document the fix. An illustrative/reference plot is not a substitute for a successful simulation.

Getting a clean WHITE plot for printing

01. With LTspice active, open Tools > Color Preferences (or Tools > Settings > Waveforms > Color Scheme in some versions).

02. Choose the Waveform/WaveForm tab, select Background, set R/G/B to 255/255/255 (white).

03. Set the first output trace to a dark navy/black and the second to dark teal or medium gray; choose dark axes/text. Increase trace width if necessary.

04. Zoom to a representative time range, show meaningful node labels and axis units, then use File > Print/Print Preview if the option is available; otherwise copy/export a readable plot screenshot and print from a document.

05. Inspect PRINT PREVIEW in grayscale: no dark background, faint text, clipped title or missing y-axis. Save PDF plus a PNG/JPG image for each trial.

Manual reference: Analog Devices LTspice Waveform Viewer Guide, "Color Control" and "Cursors". Menu wording may vary by LTspice version.

03 | HOW TO MEASURE & CALCULATE

Periodic output: Wien bridge and astable

Measure t1 and t2 at consecutive SAME-DIRECTION crossings (rising-to-rising). Then T = t2 - t1 (seconds), and f = 1/T (Hz). Use a settled waveform, not startup cycles. Percentage frequency error = 100 x |f_sim - f_theory|/f_theory.

Square-wave duty, astable

Place one cursor on a rising transition and another on the next falling transition, using the same threshold/midpoint. T_HIGH = t_fall - t_rise. Duty_HIGH (%) = 100 x (T_HIGH/T). Record T_LOW = T - T_HIGH. Ideal symmetric circuit gives about 50%, but measure actual LTspice output.

Monostable = one-shot pulse, not free-running

Measure t_start and t_end of one triggered pulse: width = t_end - t_start. If input triggers occur at 2 ms and 12 ms, the trigger period is 10 ms and trigger frequency is 100 Hz. Under one output pulse per trigger, active-pulse duty = 100 x width/trigger period. This is not an intrinsic oscillator frequency. Actual HIGH-state duty may be the complement if the active pulse goes LOW.

Bistable = state-holding comparator

No natural oscillation period, frequency or steady-state duty exists. Measure threshold-crossing/trigger time and output transition time; compare the two stable output voltages. If a pulse train toggles output, its apparent transition rate depends on stimulus, not R/C resonance. The input RC time constant R*C characterizes the coupling pulse, not a self-running cycle.

Optional automatic LTspice measurements for periodic output

Only if the output node is named OUT and crosses 0 V cleanly, add the following SPICE directives. Run again and check View > SPICE Output Log (Ctrl+L). If there are too few crossings, change RISE indices or extend stop time:

```
.meas tran rise1 WHEN V(out)=0 RISE=3

.meas tran rise2 WHEN V(out)=0 RISE=4

.meas tran Tsim PARAM rise2-rise1

.meas tran Fsim PARAM 1/Tsim

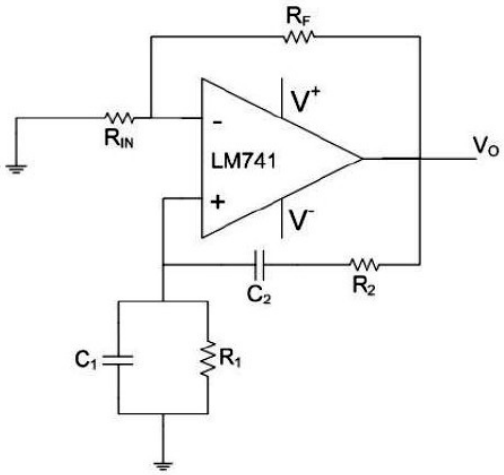
.meas tran highTime TRIG V(out) VAL=0 RISE=3 TARG V(out) VAL=0 FALL=3

.meas tran dutyHigh PARAM 100*highTime/Tsim
```

Use the highTime/dutyHigh lines only for a consistent square wave. For a sinusoid, duty-cycle figures based on zero crossing are not a useful design specification.

04 | WIEN-BRIDGE OSCILLATOR

CIRCUIT DIAGRAM: cropped directly from the user-supplied laboratory sheet



Baseline required by the laboratory handout

RIN = 1 kOhm; RF = 1.5 kOhm + 1 kOhm potentiometer; R1 = R2 = 10 kOhm; C1 = C2 = 0.1 uF. Repeated specified case: R1 = R2 = 1 kOhm, C1 = C2 = 0.22 uF. The handout says supply V+ and V- suitably.

Three trials and calculations (planned, not measured)

Trial	R1 = R2	C1 = C2	f0 theory	T0 theory
W1	10 kOhm	0.1 uF	159.15 Hz	6.283 ms
W2	1 kOhm	0.22 uF	723.43 Hz	1.382 ms
W3	4.7 kOhm	0.1 uF	338.63 Hz	2.953 ms

Theory / calculation to copy into rough record

For matched arms: f0 = 1/(2*pi*R*C), T0 = 2*pi*R*C. Ideal loop requires closed-loop gain approximately 3; actual LM741 amplitude depends on feedback, supply and model.

Run instructions for this circuit

01. Use R1 = R2 and C1 = C2 for the simple frequency formula.

02. Set feedback gain G = 1 + RF/RIN near 3 (RF near 2 kOhm for RIN 1 kOhm). Tweak RF as needed to start without severe clipping.

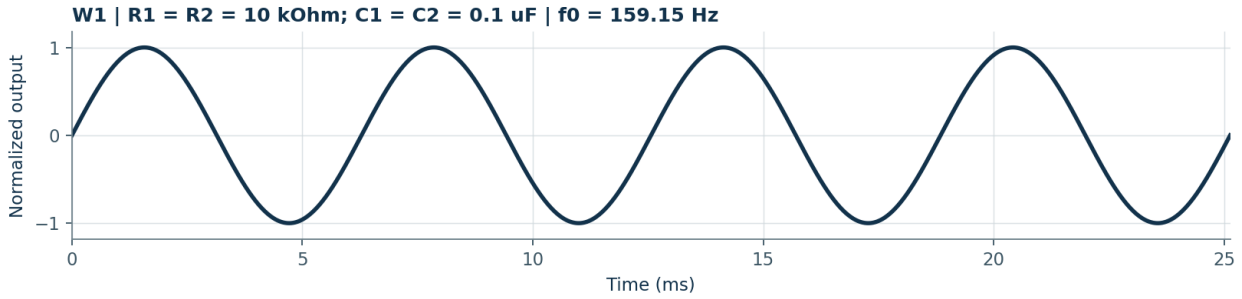
03. Observe settled sine cycles, not the initial startup transient.

Suggested starting transient directive: .tran 0 100m 0 10u (adjust for actual response; stop time and max step are not mandatory lab values).

Measure T and f from positive-going sine crossings; record Vpp if useful. Duty cycle: N/A (sinusoidal output).

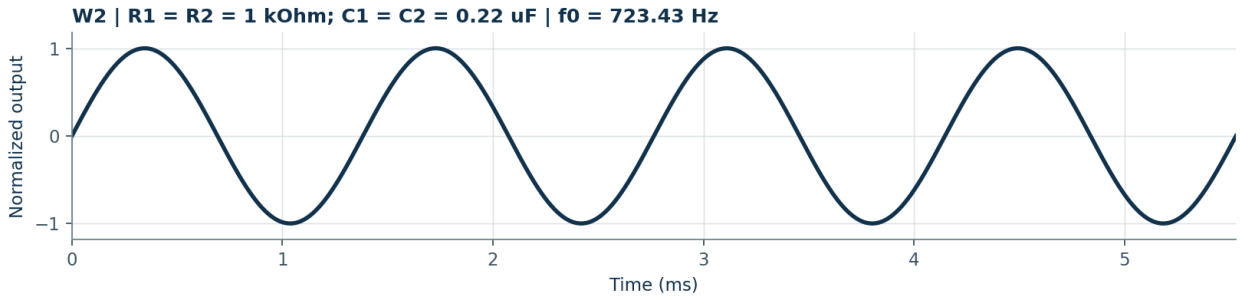
05 | WIEN-BRIDGE OSCILLATOR: REFERENCE WAVEFORMS

These are MATHEMATICAL/SCHEMATIC illustrations, not exports from LTspice. They show the expected general timing and how the graph should be presented on white paper. Do not paste them as experimental evidence.



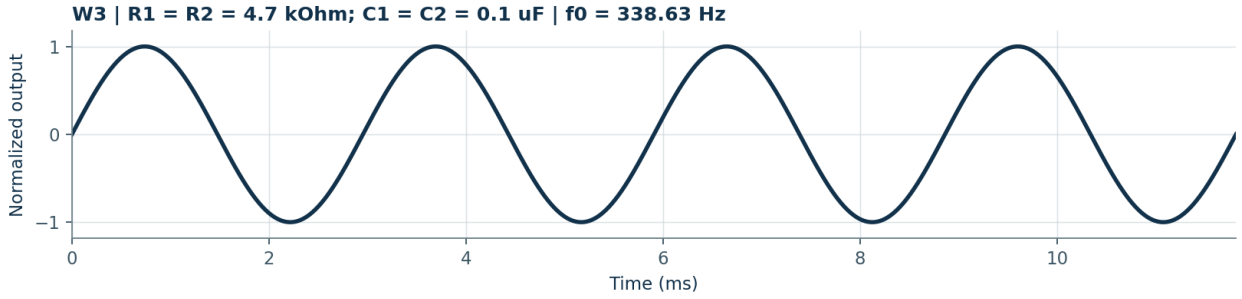
THEORETICAL SHAPE ONLY - NOT LTspice DATA

Illustration W1: use it to recognize waveform shape, then print the corresponding REAL LTspice graph.



THEORETICAL SHAPE ONLY - NOT LTspice DATA

Illustration W2: use it to recognize waveform shape, then print the corresponding REAL LTspice graph.



THEORETICAL SHAPE ONLY - NOT LTspice DATA

Illustration W3: use it to recognize waveform shape, then print the corresponding REAL LTspice graph.

06 | WIEN-BRIDGE OSCILLATOR: OBSERVATION + PRINT PAGE

Complete with actual values from the LTspice waveform cursors. A blank field does NOT mean zero. Change values only after saving the baseline trial.

Trial	R/C final	t1 (ms)	t2 (ms)	T sim (ms)	f sim (Hz)	Vpp	Error %
W1	_____	_____	_____	_____	_____	_____	_____
W2	_____	_____	_____	_____	_____	_____	_____
W3	_____	_____	_____	_____	_____	_____	_____

Three authentic waveform screenshots to paste / replace

PASTE REAL LTspice waveform here: W1 | White background + title + readable scale + actual R/C

PASTE REAL LTspice waveform here: W2 | White background + title + readable scale + actual R/C

PASTE REAL LTspice waveform here: W3 | White background + title + readable scale + actual R/C

Before printing, record here: simulator version _____; LM741 model _____; supply rails _____; any changed setup / corrections _____

Student / friend check: [ ] Correct schematic [ ] Three values [ ] Real screenshots [ ] Axes/labels [ ] Measured numbers [ ] White plot [ ] Rough record calculation

07 | ASTABLE MULTIVIBRATOR

CIRCUIT DIAGRAM: cropped directly from the user-supplied laboratory sheet

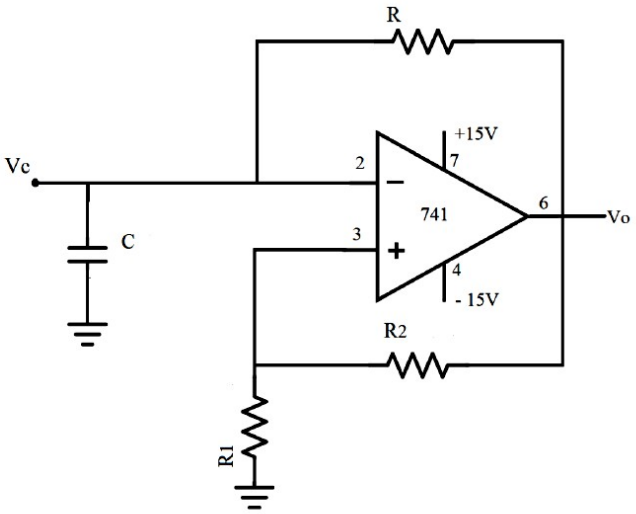


Figure 2 Astable Multivibrator

Baseline required by the laboratory handout

R1 = R2 = 10 kOhm; R = 4.7 kOhm; C = 0.1 uF; 741 supply +15 V / -15 V per lab figure. Use potentiometer or trial resistor for R.

Three trials and calculations (planned, not measured)

Trial	R	C	T theory	f theory	Duty ideal
A1	4.7 kOhm	0.1 uF	1.0327 ms	968.34 Hz	50%
A2	10 kOhm	0.1 uF	2.1972 ms	455.12 Hz	50%
A3	4.7 kOhm	0.047 uF	0.4854 ms	2060.30 Hz	50%

Theory / calculation to copy into rough record

beta = R1/(R1+R2) = 0.5; t_HIGH = t_LOW = R*C*ln((1+beta)/(1-beta)) = R*C*ln(3); T = 2*R*C*ln(3); f = 1/T. Assumes symmetric output saturation.

Run instructions for this circuit

01. Plot V(out) and VC simultaneously. The capacitor follows exponential charge/discharge between Schmitt thresholds.

02. Start with the exact handout values; vary R (or C) while R1/R2 remain unchanged.

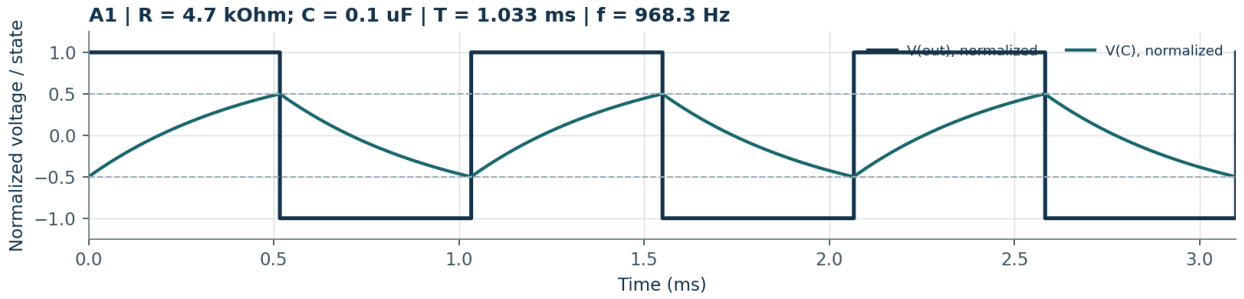
03. Measure HIGH and LOW times separately; ideal duty is 50%.

Suggested starting transient directive: .tran 0 30m 0 2u (adjust for actual response; stop time and max step are not mandatory lab values).

T = T_HIGH + T_LOW; f = 1/T; duty = 100*T_HIGH/T. Compare theory with LTspice.

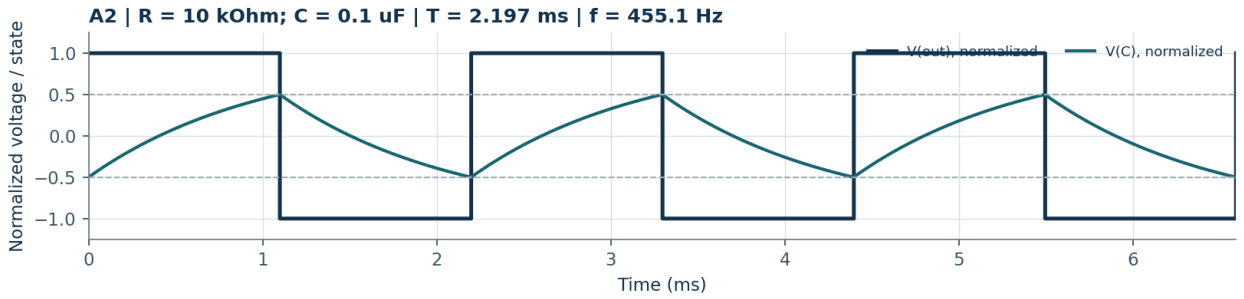
08 | ASTABLE MULTIVIBRATOR: REFERENCE WAVEFORMS

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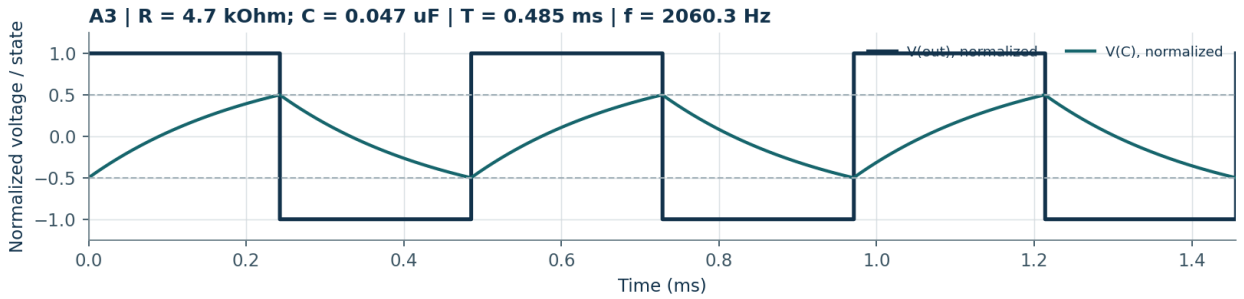
IDEAL SYMMETRICAL EXAMPLE - NOT LTspice DATA

Illustration A1: use it to recognize waveform shape, then print the corresponding REAL LTspice graph.



IDEAL SYMMETRICAL EXAMPLE - NOT LTspice DATA

Illustration A2: use it to recognize waveform shape, then print the corresponding REAL LTspice graph.



IDEAL SYMMETRICAL EXAMPLE - NOT LTspice DATA

Illustration A3: use it to recognize waveform shape, then print the corresponding REAL LTspice graph.

09 | ASTABLE MULTIVIBRATOR: OBSERVATION + PRINT PAGE

Complete with actual values from the LTspice waveform cursors. A blank field does NOT mean zero. Change values only after saving the baseline trial.

Trial	R/C final	T HIGH (ms)	T LOW (ms)	T sim (ms)	f sim (Hz)	Duty %
A1	_____	_____	_____	_____	_____	_____
A2	_____	_____	_____	_____	_____	_____
A3	_____	_____	_____	_____	_____	_____

Three authentic waveform screenshots to paste / replace

PASTE REAL LTspice waveform here: A1 | White background + title + readable scale + actual R/C

PASTE REAL LTspice waveform here: A2 | White background + title + readable scale + actual R/C

PASTE REAL LTspice waveform here: A3 | White background + title + readable scale + actual R/C

Before printing, record here: simulator version _____; LM741 model _____; supply rails _____; any changed setup / corrections _____

Student / friend check: [ ] Correct schematic [ ] Three values [ ] Real screenshots [ ] Axes/labels [ ] Measured numbers [ ] White plot [ ] Rough record calculation

10 | MONOSTABLE MULTIVIBRATOR

CIRCUIT DIAGRAM: cropped directly from the user-supplied laboratory sheet

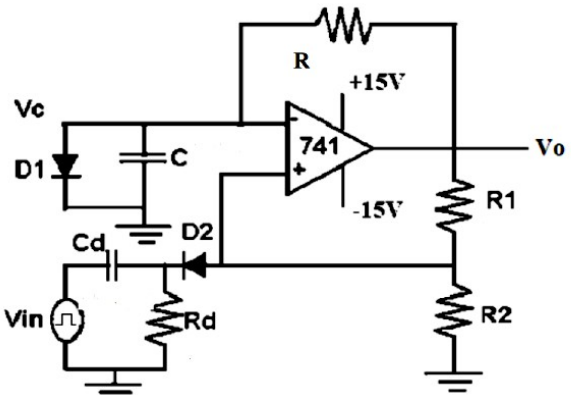


Figure 3 Monostable Multivibrator

Baseline required by the laboratory handout

R1 = R2 = 10 kOhm; Rd = 470 Ohm; C = 0.1 uF; Cd = 0.01 uF; R = 15 kOhm; 741 supply +15 V / -15 V per lab figure. Diodes D1 and D2 are essential.

Three trials and calculations (planned, not measured)

Trial	R	C	tp ideal approx	Trigger rate	Active duty ideal
M1	15 kOhm	0.1 uF	1.040 ms	100 Hz*	10.40%*
M2	10 kOhm	0.1 uF	0.693 ms	100 Hz*	6.93%*
M3	15 kOhm	0.22 uF	2.287 ms	100 Hz*	22.87%*

Theory / calculation to copy into rough record

General first-order timing: tp = R*C*ln[(Vstart - Vfinal)/(Vthreshold - Vfinal)] using the actual reset/clamp and switching levels. Common ideal approximation for beta = 0.5 and negligible diode drop: tp ~= R*C*ln(2). Actual diode drops and op-amp limits change tp.

Run instructions for this circuit

01. Include an external input pulse source and plot BOTH input trigger and output.

02. Example trigger spacing: 10 ms (100 Hz). Verify actual polarity/edge that fires the one-shot from the diode network.

03. Keep enough spacing for full return to the stable state; changing R/C changes output pulse width, not an autonomous frequency.

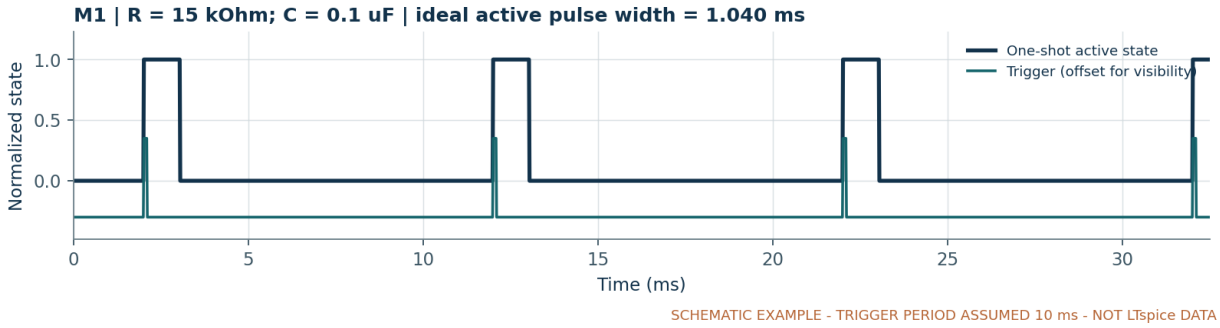
Suggested starting transient directive: .tran 0 40m 0 2u (adjust for actual response; stop time and max step are not mandatory lab values).

Suggested starting trigger for LTspice source (check polarity against diode D2): PULSE(0 -5 1m 1u 1u 0.1m 10m). Plot trigger and V(out); if no pulse, inspect actual threshold, then correct the trigger polarity/amplitude rather than inventing a waveform.

Record measured pulse width. With one pulse per 10 ms trigger cycle, f_output repetition = 100 Hz (set by external trigger), and active duty = 100*tp/10 ms. Do not call this an intrinsic oscillator frequency.

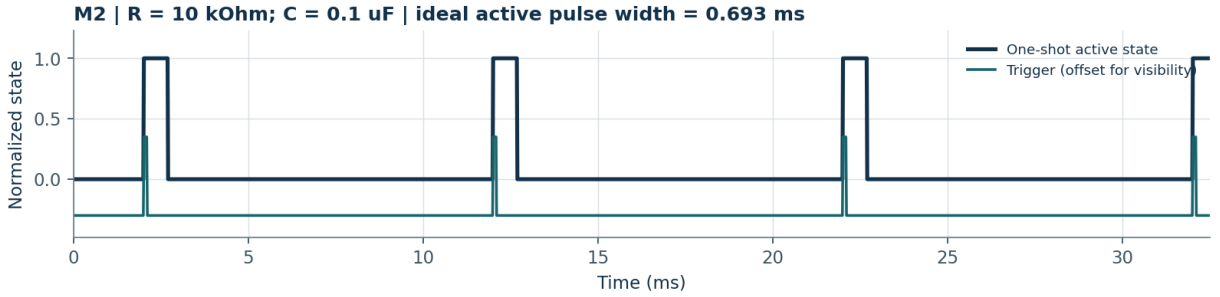
11 | MONOSTABLE MULTIVIBRATOR: REFERENCE WAVEFORMS

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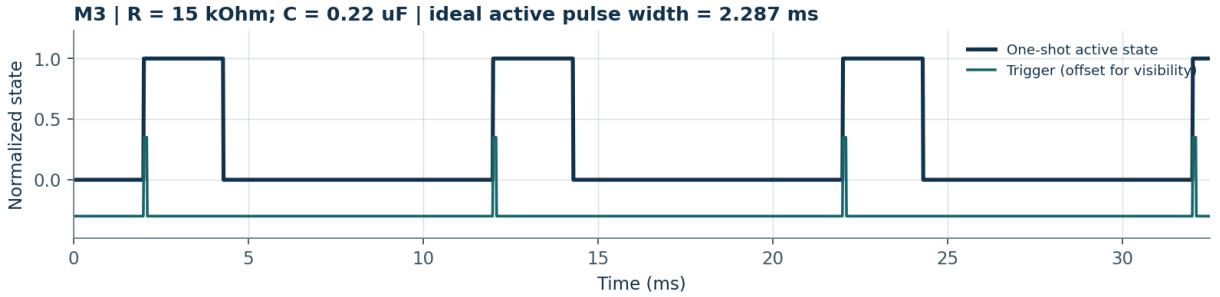
SCHEMATIC EXAMPLE - TRIGGER PERIOD ASSUMED 10 ms - NOT LTspice DATA

Illustration M1: use it to recognize waveform shape, then print the corresponding REAL LTspice graph.



SCHEMATIC EXAMPLE - TRIGGER PERIOD ASSUMED 10 ms - NOT LTspice DATA

Illustration M2: use it to recognize waveform shape, then print the corresponding REAL LTspice graph.



SCHEMATIC EXAMPLE - TRIGGER PERIOD ASSUMED 10 ms - NOT LTspice DATA

Illustration M3: use it to recognize waveform shape, then print the corresponding REAL LTspice graph.

12 | MONOSTABLE MULTIVIBRATOR: OBSERVATION + PRINT PAGE

Complete with actual values from the LTspice waveform cursors. A blank field does NOT mean zero. Change values only after saving the baseline trial.

Trial	R/C final	Trig period (ms)	tp sim (ms)	Trig Hz	Active duty %	Stable level
M1	_____	_____	_____	_____	_____	_____
M2	_____	_____	_____	_____	_____	_____
M3	_____	_____	_____	_____	_____	_____

Three authentic waveform screenshots to paste / replace

PASTE REAL LTspice waveform here: M1 | White background + title + readable scale + actual R/C

PASTE REAL LTspice waveform here: M2 | White background + title + readable scale + actual R/C

PASTE REAL LTspice waveform here: M3 | White background + title + readable scale + actual R/C

Before printing, record here: simulator version _____; LM741 model _____; supply rails _____; any changed setup / corrections _____

Student / friend check: [ ] Correct schematic [ ] Three values [ ] Real screenshots [ ] Axes/labels [ ] Measured numbers [ ] White plot [ ] Rough record calculation

13 | BISTABLE MULTIVIBRATOR

CIRCUIT DIAGRAM: cropped directly from the user-supplied laboratory sheet

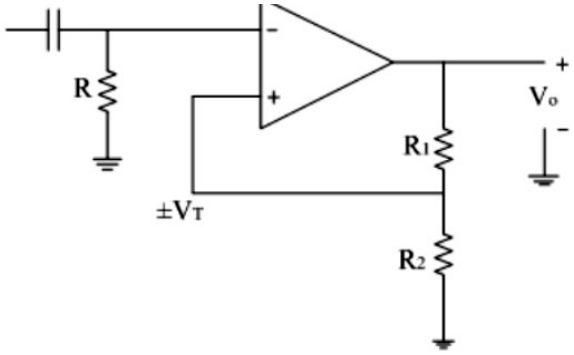


Figure 4 Bistable Multivibrator

Baseline required by the laboratory handout

R1 = R2 = 10 kOhm; R = 100 Ohm; C = 0.1 uF. Page 3 figure shows AC-coupled trigger into inverting pin; positive feedback to noninverting pin.

Three trials and calculations (planned, not measured)

Trial	R	C	Input RC tau	Theoretical f	Duty
B1	100 Ohm	0.1 uF	10.00 us	N/A	N/A
B2	220 Ohm	0.1 uF	22.00 us	N/A	N/A
B3	100 Ohm	0.047 uF	4.70 us	N/A	N/A

Theory / calculation to copy into rough record

beta = R2/(R1+R2) = 0.5 (output feedback divider); switching thresholds approximately +/-beta*Vsat, where Vsat is the actual modeled output saturation level. Trigger-coupling time constant tau = R*C (NOT output period).

Run instructions for this circuit

01. Apply distinct trigger edges through series input capacitor and show both stable output levels.

02. Capacitor creates decaying edge pulses; ensure input edges actually cross the Schmitt switching thresholds without exceeding model/input ratings.

03. After each trigger output must remain at a state until the opposite qualifying trigger; it does not free-run.

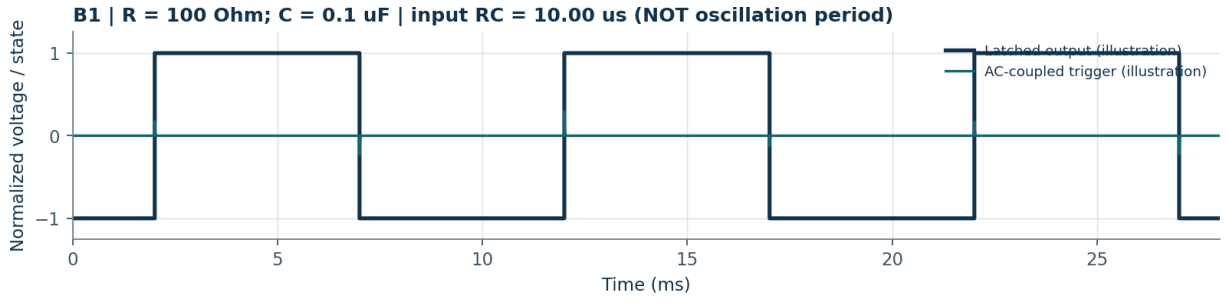
Suggested starting transient directive: .tran 0 30m 0 200n (adjust for actual response; stop time and max step are not mandatory lab values).

Suggested exploratory trigger: PULSE(-7 7 1m 1u 1u 5m 10m), through the capacitor as drawn. This is a trial input, NOT a lab-specified voltage; check op-amp model/input limits and confirm both transitions before reporting.

Record successful SET/RESET transitions, held HIGH and LOW voltages, threshold estimate and switching delay if readable. Period, frequency and duty cycle = N/A: no free-running oscillation.

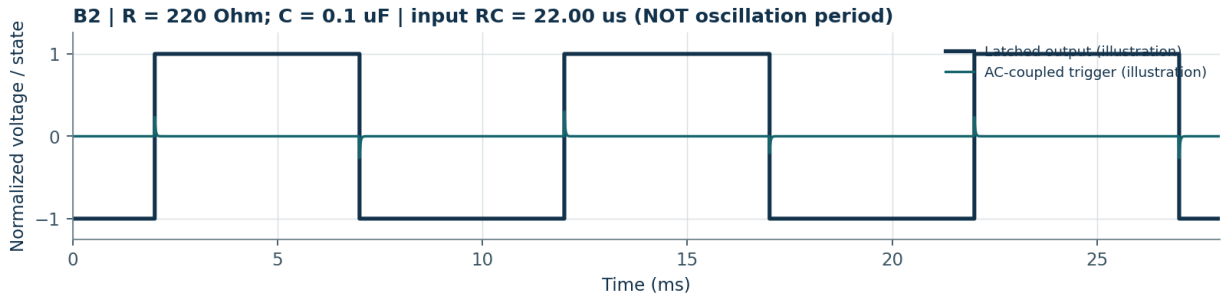
14 | BISTABLE MULTIVIBRATOR: REFERENCE WAVEFORMS

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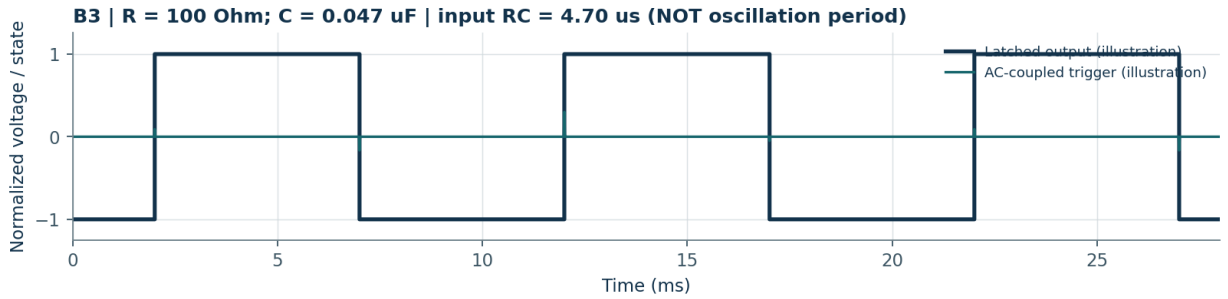
QUALITATIVE EDGE-TRIGGERED EXAMPLE - NOT LTspice DATA

Illustration B1: use it to recognize waveform shape, then print the corresponding REAL LTspice graph.



QUALITATIVE EDGE-TRIGGERED EXAMPLE - NOT LTspice DATA

Illustration B2: use it to recognize waveform shape, then print the corresponding REAL LTspice graph.



QUALITATIVE EDGE-TRIGGERED EXAMPLE - NOT LTspice DATA

Illustration B3: use it to recognize waveform shape, then print the corresponding REAL LTspice graph.

15 | BISTABLE MULTIVIBRATOR: OBSERVATION + PRINT PAGE

Complete with actual values from the LTspice waveform cursors. A blank field does NOT mean zero. Change values only after saving the baseline trial.

Trial	R/C final	Input +/- edges	VHIGH	VLOW	Changed state?	Delay (if measurable)
B1	_____	_____	_____	_____	_____	_____
B2	_____	_____	_____	_____	_____	_____
B3	_____	_____	_____	_____	_____	_____

Three authentic waveform screenshots to paste / replace

PASTE REAL LTspice waveform here: B1 | White background + title + readable scale + actual R/C

PASTE REAL LTspice waveform here: B2 | White background + title + readable scale + actual R/C

PASTE REAL LTspice waveform here: B3 | White background + title + readable scale + actual R/C

Before printing, record here: simulator version _____; LM741 model _____; supply rails _____; any changed setup / corrections _____

Student / friend check: [ ] Correct schematic [ ] Three values [ ] Real screenshots [ ] Axes/labels [ ] Measured numbers [ ] White plot [ ] Rough record calculation

16 | ROUGH RECORD: COPY-READY WRITING PLAN

Title / Aim

Experiment 8 - Oscillator and Multivibrator Circuits. (i) To design and study the Wien-bridge oscillator. (ii) To design and study astable, monostable and bistable multivibrator circuits using an operational amplifier.

Apparatus / software

LTspice, LM741 op-amp macromodel, resistors, capacitors, diodes for monostable circuit, voltage/pulse sources and positive/negative power supplies.

General procedure

01. Draw the four circuits exactly as in the laboratory handout (Figures 1-4) and select specified initial values.

02. Use suitable +/- supplies for 741; for multivibrators use +/-15 V as shown. Check model connection and initial conditions.

03. Set transient analysis, run the circuit, and observe V(out) with V(C) or V(in) as applicable.

04. Make three trials per circuit with planned R/C variations and save each separately.

05. Use cursors to measure the time interval and output levels. Calculate the relevant quantity using the formula for that circuit.

06. Obtain readable light-background LTspice printouts; note observed values and compare with calculated values.

Sample worked substitution - Wien bridge

W1: R1 = R2 = 10 kOhm, C1 = C2 = 0.1 uF. f0 = 1/[2*pi*(10,000)*(0.1 x 10^-6)] = 159.155 Hz. T0 = 1/f0 = 6.283 ms. Now replace R/C with W2 and W3 values; use actual f_sim for error.

Sample worked substitution - astable

A1: beta = 10k/(10k+10k) = 0.5. t_HIGH = 4.7k*0.1u*ln(3) = 0.516 ms; T = 2*t_HIGH = 1.033 ms; f = 1/T = about 968.4 Hz; ideal duty = 50%.

Sample worked substitution - monostable

M1: beta = 0.5, R = 15k, C = 0.1u. Under the ideal reset/clamp approximation, tp ~ 15k*0.1u*ln(2) = 1.040 ms. If retriggered every 10 ms, theoretical active-state duty is about 10.40%. Compare with the actual diode+741 simulation.

Sample worked substitution - bistable

B1: beta = 10k/(10k+10k) = 0.5, so thresholds are ~+/-0.5 Vsat (measure Vsat first). The input RC time constant is 100 Ohm*0.1uF = 10 us. This is NOT a waveform period. Show state changes and report frequencies as N/A.

17 | RESULTS, CAUTIONS & FINAL CHECK

Result wording (fill observed values first)

The Wien-bridge oscillator produced a [measured shape] waveform with measured frequencies W1 = _____ Hz, W2 = _____ Hz, W3 = _____ Hz. The astable multivibrator produced a square-wave output with measured frequencies A1 = _____ Hz, A2 = _____ Hz, A3 = _____ Hz and observed duties _____ / _____ / _____ %. The monostable generated trigger-controlled pulses with widths _____ / _____ / _____ ms. The bistable was observed switching and holding two stable states under trigger input [YES/NO]. The measured values [agree/do not agree] with ideal calculations within the simulation model's limitations.

Troubleshooting notes for the friend

- Sine output is a flat line: check correct LM741 supplies/model mapping, feedback sign, startup noise or initial condition, loop gain ~3, and simulation stop time.
- Sine is clipped: reduce gain slightly / permit realistic amplitude control; do not claim it is a clean sine if saturated.
- Astable no oscillation: check positive-feedback node polarity, R-C connection, ground reference, +/- rails and initial conditions.
- Monostable does not respond: plot input, diode orientation, and noninverting threshold, then test the correct trigger edge/polarity; ensure recovery before next pulse.
- Bistable not toggling: the AC-coupled input edge may be below switching threshold. Inspect the actual V(-) pulse and +/-beta*Vsat levels; change trigger input within safe modeled limits, not blindly.
- Automatic measurements fail: wrong node name, too few rising crossings, startup transient, trace not crossing 0V, or window too short. Use manual cursors.

Printer checklist

Check	Required confirmation
Circuit completeness	4 circuit diagrams and related written steps
Trial coverage	W1-W3, A1-A3, M1-M3, B1-B3 (12 total)
Screenshots	All are ACTUAL LTspice traces, not reference sketches
Print style	White background, dark traces, scales/units visible
Math	T and f for oscillators; duty for astable; tp and trigger rate for mono; state levels for bi
Records	Fair: stick printouts. Rough: write experiment/calculations.
Simulation integrity	Every .asc saved and every observation traceable to a trial

References / notes

1. Supplied EC3091D Experiment-8 lab handout, pages 1-3 (circuit diagrams and baseline values).

2. Teacher message supplied by student: print actual waveforms in fair record; write experiment in rough record; vary R/C for 2-3 plots/circuit and record timing parameters.

3. Analog Devices LTspice Waveform Viewer Guide (color preferences, dual cursors, .meas, plots): https://analogdevicesinc.github.io/ltspice-reference/ai_ref/WAVEFORM-VIEWER-GUIDE.html

4. Op-amp astable timing equation uses ideal symmetric thresholds. Monostable timing is an explicit approximation dependent on real diode clamp and output swing; actual values must be measured.